Modeling Selection in Active Cross-situational Word Learning

George Kachergis (kachergis@stanford.edu)
Department of Psychology, Stanford University
Stanford, CA

Martin Zettersten (mzettersten@ucsd.edu)
Department of Cognitive Science, University of California, San Diego
San Diego, CA

Abstract

Word learning is a complex, interactive process, in which learners do not passively observe a stream of words and referents, but are rather actively involved in selecting referents and directing their attention – both of which are likely determined by their current knowledge state. Thus, understanding how learners actively select information during word learning can reveal the cognitive mechanisms underlying language acquisition. The present study reanalyzes data from Zettersten & Saffran (2021), in which adults and children (ages 3-8) learned novel word-referent mappings through cross-situational learning, then selected which referents to receive additional training on. Participants' individual training data, selection choices, and test trials were used to fit four associative word learning models, and to infer the most likely of five sampling strategies. Models with a bias to attend to stimuli with uncertain (i.e. high-entropy) knowledge states best accounted for the data, though there were individual differences in the degree to which this bias appeared to be the main driver of learning. Approximately half of participants appear to be driven solely by this bias (51%), while another 48% were best accounted for by a model additionally engaging a competing familiarity bias. Analysis of selection strategies revealed both developmental consistencies and differences: while adults and children predominantly use an information gain strategy to reduce uncertainty about ambiguous items, adults deployed more diverse strategies (confirmatory, margin, and random sampling). These findings suggest that reducing uncertainty can drive sampling behavior even in young children, but that the repertoire of sampling strategies grows with maturity. Moreover, individual differences in learning mechanisms—particularly learning rate and memory—are more predictive of word learning success than sampling strategy alone.

Keywords: active learning; cross-situational word learning; self-directed learning; uncertainty reduction

Introduction

Early word learning is often viewed as largely determined by the learner's linguistic environment, where acquisition is constrained by what is present in the input. Increasingly, however, research reveals that learners–even young children–actively select and shape their input, and that such self-directed learning can accelerate acquisition (Smith, Jayaraman, Clerkin, & Yu, 2018; Twomey & Westermann, 2018). Understanding the strategies learners use to curate their experience can advance theories of language acquisition, reveal the cognitive mechanisms and capacities available across development, and inform the design of educational curricula that match learners' expanding capabilities.

Despite growing interest in active learning, most computational models of word learning have been evaluated exclusively on passive cross-situational learning paradigms, in which participants observe ambiguous training trials presenting multiple word-referent pairs without indication of the correct mappings (Kachergis, Yu, & Shiffrin, 2016; Yu & Smith, 2007). While model comparisons in these passive contexts are informative, active learning paradigms in which learners control some aspect of their input (e.g., selecting which item to learn about next) can be more diagnostic of underlying learning mechanisms and may better capture individual differences in learning success (Gureckis & Markant, 2012; Markant, Settles, & Gureckis, 2016). Yet few active word learning experiments have been conducted, and fewer still have attempted to explain active learning behavior with computational models (though see Kachergis, Yu, & Shiffrin, 2013).

The goal of the current study is to test how well associative models can explain both learning outcomes and information-seeking during cross-situational word learning, identify what sampling strategies learners pursue, and track how sampling strategies develop with age. To investigate these questions, we model data from a set of active, cross-situational word learning experiments with children and adults (Zettersten & Saffran, 2021). In these experiments, adults preferentially selected items with ambiguous word-referent mappings during a sampling phase, while children (ages 3-8) increased their preference to sample ambiguous items across age. We address three main questions: First, which model architecture best captures individual learning performance? Second, do specific model parameters (learning rate, memory decay, attention to uncertainty) predict learning success? To address these first two issues, we chose a family of models that instantiate trial-by-trial associative learning with memory decay and selective attention to familiar and/or uncertain (or novel) stimuli, which have been shown to account well for semantic bootstrapping and mutual exclusivity effects in prior work (Kachergis, Yu, & Shiffrin, 2012, 2016). Third, what sampling strategies (e.g., entropy reduction vs. confirmatory sampling) best explain participants' information-seeking choices, and how do these strategies develop with age?

Understanding children's and adults' sampling strategies during word learning has important implications for theories of vocabulary development. Past work shows that both children and adults actively seek information about word meanings (Hembacher, DeMayo, & Frank, 2020), which may help explain rapid vocabulary growth (Hidaka, Torii, & Kachergis, 2017). However, we lack a formal understanding of the specific decision rules governing when learners seek new information versus consolidate existing knowledge (Settles, 2012; though see Gelderloos, Mahmoudi Kamelabad, & Alishahi, 2020; Keijser, Gelderloos, & Alishahi, 2019). Recent developmental work documents broad changes in information-seeking strategies across childhood in domains such as causal learning (Nussenbaum et al., 2019) and foraging (Meder, Wu, Schulz, & Ruggeri, 2021; Nussenbaum et al., 2023), suggesting that strategic sampling may emerge gradually with age and cognitive development. The current work therefore explores what sampling decision rules best explain children's and adults' information-seeking during word learning, and how learning mechanisms relate to sampling strategies.

A final important theoretical question concerns the relationship between learning success and sampling strategy. Zettersten & Saffran (2021) found that adults who sampled more ambiguous items during the selection phase showed higher test accuracy. However, the directionality of this relationship remains unclear: does sampling ambiguous items *cause* better learning, or does successful learning *enable* more sophisticated ambiguity-seeking strategies? Their supplementary analyses suggested the latter: as participants became more successful at learning during training, they became more likely to seek information about ambiguous items. We investigate this question by examining whether individual differences in learning parameters (learning rate, memory, attention biases) predict both test performance and sampling strategy, potentially illuminating the mechanisms linking learning capacity to active information-seeking.

Method

Data

All data were taken from Zettersten & Saffran (2021). Data from Experiment 1 (31 adults), Experiment 2 (40 children, 4-9 years of age), and adults in the Ambiguous Condition of Experiment S1 (58 adults) were pooled, as they all had the same design, except for an additional test before the selection phase in Exp. S1. The training phase of these data contained 24 trials, each comprised of 2 referents and 2 words. The particular words and referents involved in each pairing were randomized per participant, but the same training trial sequence–and thus co-occurrence statistics–were experienced by all participants.

Experiment 3, with data from 58 children, 3 to 8 years of age, had a similar design, except that two of the word-referent pairs were already familiar to children ("dog" and "pig").

Selection trials for kids differ slightly from adults in that there was no random label-referent pair given: thus, the evidence children received after selection was unambiguous, while adults received a training trial with a randomly-selected label and its corresponding referent. The models will see what the learner saw: for children, that would be an unambiguous trial with the selected referent and its label; while for adults, the model will be presented with the selected stimulus pair alongside the random pair the learner saw.

Models

We focused on analyzing data from the familiarity- and uncertainty-biased associative model (Kachergis et al., 2012, 2016), and variations of that model, described below. We chose not to test any hypothesis-testing (HT) model as they are not consistent with the adult data in Zettersten & Saffran (2021): for ambiguous items in the training phase, there was never any disconfirming evidence, so a hypothesis-tester's initial guess (i.e. hypothesis) for each label would be confirmed on subsequent appearances, and there would be no reason for (or basis for) preferentially selecting ambiguous referents during the sampling phase, which is what adults were observed to do. While it is possible that children – who were at chance for selecting ambiguous vs. unambiguous referents – are employing a hypothesis-testing strategy, some of the selection strategies we want to evaluate across development are only possible to use if a learner is simultaneously entertaining multiple hypotheses for a given label – equivalent to associations rather than the mutually-exclusive hypotheses favored in HT accounts (e.g. Woodard, Gleitman, & Trueswell, 2016).

Familiarity- and Uncertainty-biased Associative Model (FUAM) The FUAM assumes that learners accumulate associations between all presented words and referents on each trial, but that their attention is biased more towards particular word-referent associations on the basis of prior experience. In particular, more attention (i.e. associative weight) is directed to associations between words and referents that have appeared together on earlier trials (i.e., a familiarity bias). In addition, more attention is directed to stimuli that have higher uncertainty (i.e. entropy, H) in their associations, whether because they are novel (and thus have no pre-existing associations) or because they have diffuse associations with several different stimuli. The model assumes learners knowledge grows trial-by-trial, and that a fixed amount of associative weight (learning rate χ) is distributed among the presented word-referent associations on a given trial in proportion to the prior familiarity of the association and the uncertainty of the involved word and referent. The relative strength of the familiarity and uncertainty bias is governed by a parameter λ: for values of $\lambda > 1$ the model focuses more attention on uncertain stimuli, while for values of $\lambda < 1$ the model focuses more attention on familiar word-referent associations. Associations are also assumed to decay from trial to trial, at a rate governed by memory fidelity parameter α. The update rule for adjusting the association $M_{w,o}$ between word w and referent o on a given trial is

$$M_{w,o} = \alpha M_{w,o} + \frac{\chi \cdot e^{\lambda \cdot (H(w)+H(o))} \cdot M_{w,o}}{\sum_{w \in W} \sum_{o \in O} e^{\lambda \cdot (H(w)+H(o))} \cdot M_{w,o}} \quad (1)$$

When tested on word w, the model assumes learners choose referent o from the offered alternatives in proportion to their associative strength $M_{w,o}$ (i.e., Luce choice).

The other models we tested are variations on this model.

Familiarity-biased Associative Model (FAM) The familiarity-biased associative model (FAM) implemented just the familiarity bias of the FUAM, without the uncertainty bias term, and thus only implements a bias for already-strong associations, using learning rate (χ) and decay (α) parameters.

Uncertainty-biased Associative Model (UAM) The uncertainty-biased associative model (UAM) implemented only the uncertainty bias of the FUAM, without the familiarity bias term, and uses the same three parameters as the FUAM.

Novelty-biased Associative Model (NAM) The novelty-biased associative model (NAM), like the UAM, has no familiarity bias, but instead of the entropy terms ($H(w)$ and $H(o)$) uses novelty (e.g., $1/(frequency(w)+1)$).

Fitting Procedure

We first fitted each model to each participant's data individually, finding parameter values per participant to maximize the log-likelihood of the model's choices with respect to theirs. We fitted each model to just the passive training trials, as well as to the passive training trials and the participant's selection trials, to confirm that including the selection trials yielded a better fit. We will report each model's average fit, and select the best-fitting model overall for further analysis.

Modeling Sampling Choices

Next, we used the best-fit models of participants' learning to investigate their selection strategies in the sampling phase of the experiment. We considered three sampling decision rules commonly attested in the information-seeking literature to explain active choices: maximizing entropy reduction (information gain) and margin sampling.

Maximizing entropy reduction. In this decision rule, the model's goal is to maximally reduce the entropy of the object-label probability distribution across objects (Nelson, 2005; Settles, 2012). Sampling choices are modeled based on comparing the relative entropy reduction provided by making a given sampling selection of a referent s_o (across the labels w):

$$\max_{s_o,w}(H(w)-H(w|s_o)) \tag{2}$$

Modeling selections were made by determining, on any given trial for a participant, which object maximally reduced entropy (averaging across all possible random distractors that could be chosen in the case of the adult data).

Margin sampling. An alternative strategy documented in describing active sampling strategies is focusing on reducing uncertainty for a more constrained set of items (Markant et al., 2016). This strategy can be formalized as *margin sampling*, in which the model focuses only on the boundary between two alternative options. Margin sampling predicts that learners will focus on instances where there is highest ambiguity between the two most likely labels for a given referent. Formally, if the distribution of labels ($\{l_1, l_2, ..., l_n\}$) for a given referent o are ordered from most (p_1) to least (p_n) likely, $\{p_1, p_2, ..., p_n\}$, the margin between labels is given by

$$LM(x) = 1-(p_1-p_2) \tag{3}$$

In other words, the model selects the object that has the greatest ambiguity or competition between its two strongest associates.

Alternative strategies. We also tested a few additional strategies. First, we also considered a *maximum certainty* (or confirmatory) strategy. Learners may gravitate towards confirming choices they already have high certainty for (Wason, 1960; Zettersten, Cutler, & Lew-Williams, 2023). Second, we considered a *novelty-based* strategy in which the model based its choices on the relative novelty (or familiarity) of the choice options. Third, we consider a *random* selection strategy as a baseline comparison.

Results

Model	N best fit	Fam	Unc	Nov	FUAM
FUAM	67	1.66	1.66	1.65	1.42
Fam	24	2.00	2.09	2.00	2.00
Nov	3	3.22	3.42	3.01	3.19
Unc	96	3.56	2.57	3.49	3.37

Table 1: Number of participants best fitted per model, and average negative log-likelihood (lower=better) per model.

For all four models, including the selection trials yielded a better fit to the data, showing that it is not unreasonable to apply these models to the selection trials, and supporting the idea that the model's knowledge state and parameters may bear some relation to participants' selection strategies. For model comparison, we calculated BIC (Bayesian Information Criterion), defined as $BIC = ln(N)k - 2ln(L)$, where L is the maximized likelihood, k is the number of parameters, and N is the number of data points (here, $N = 1712$ test trials). The model with the lowest BIC is selected, as it balances model complexity and fit to the data. The familiarity-biased model yielded $BIC = 1036.17$, the novelty-biased model scored $BIC = 1026.68$, and the familiarity- and uncertainty-biased model (FUAM) had $BIC = 975.22$. The uncertainty-biased model (UAM) yielded the best fit, with $BIC = 859.09$. Recognizing that different participants may be operating under different learning regimes, we opted to investigate how many participants were best fitted by each model. The uncertainty-biased model best matched behavior of 96 of the 190 participants (51% of the sample), and the average fit (negative log-likelihood; see Table 1) of the model to its best-fitted participants was much better (i.e. lower) than any other model (2.57 vs. 3.37 by the FUAM model). The FUAM achieved the best fit for 67 participants, with the best average fit to that group seen in Table 1 (1.42). The FUAM also matched the best fit of the familiarity-biased model to 3 decimal places (see Table 1) for the 24 participants that were best fitted by that model, making for a total of 91 participants (48% of the sample) that are best or nearly equally well accounted for by the FUAM model (the uncertainty model did not fit as well to the strength participants). The novelty-biased model best fitted only 3 participants, although it's worth noting that its average fit to the strength participants was nearly the same as the FUAM and strength models. We will focus our remaining analyses on the model parameters for the two best-fitting models, the familiarity- and uncertainty-biased associative model (FUAM) and its cousin the uncertainty-biased associative model (UAM).

Model parameters

The FUAM and UAM models achieved similar fit ($|NLL_{FUAM} - NLL_{UAM}| \leq 0.1$) for 98 participants. The UAM achieved better fit than the FUAM ($NLL_{FUAM} - NLL_{UAM} > 0.1$) for 56 participants, and the FUAM achieved better fit than the UAM for the remaining 36 participants. We will call these participants the *Either*, *UAM*, and *FUAM* groups, respectively.

Group	N	chi	lambda	alpha	Mean accuracy
Either	98	0.32	5.34	0.77	0.59
FUAM	36	0.41	4.97	0.77	0.65
UAM	56	0.26	8.76	0.69	0.57

Table 2: Average parameter values per group of participants best fitted by each model.

Table 2 shows the average parameter values for each group of participants, and the mean accuracy and number of participants per group. None of these values are strikingly different than the overall average parameter values (mean $\chi = 0.32$, mean $\lambda = 6.28$, mean $\alpha = 0.75$), though the UAM group shows lower learning rate ($\chi = 0.26$), lower memory fidelity ($\alpha = 0.69$), and a higher focus on uncertainty ($\lambda = 8.76$). The FUAM group has a somewhat higher learning rate ($\chi = 0.41$ and the greatest overall accuracy ($M_{acc} = 0.65$).

Figure 1 shows the correlations between participants' accuracy, best-fitting model parameters, and log(age). All correlations are significant (e.g., λ vs log(age) $r = 0.14$, $t(187) = 1.99$, $p < .05$) except λ vs. χ. Most notable is that accuracy is strongly related to higher learning rate χ ($r = 0.53$) and lower memory fidelity ($\alpha = 0.58$; better learning is achieved by forgetting early, ambiguous evidence – as has been seen in other applications of this model), and only weakly related to *lambda* ($r = 0.23$) and age ($r = 0.29$). Figure 3 shows participants' best-fitting model parameters by age, revealing developmental increases in learning rate (χ) and attention to uncertainty (λ), and decrease in memory fidelity (α). Children show greater individual differences, but on average match adults' parameter values by age 8. A targeted analysis found that children (age<18) are more likely than adults to be familiarity-focused ($\lambda < 1$) rather than uncertainty-focused ($\chi^2 = 7.41$): of 22% of adults had $\lambda < 1$, while 41% of children did.

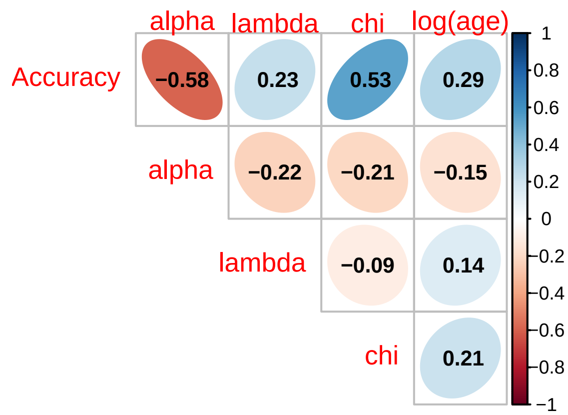

Figure 1: Correlation of individuals' best-fitting model parameters with age and test accuracy. Accuracy is strongly related to higher learning rate (chi) and lower memory fidelity (alpha).

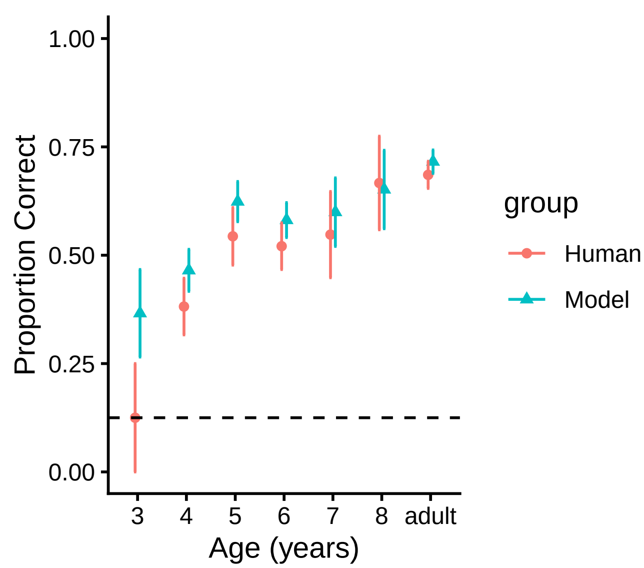

Figure 2: Participants' vs. model accuracy at test by age. The model generally matches participants' performance, except for outperforming 3-year-olds, who performed at chance (dotted line). Error bars show +/-1SE.

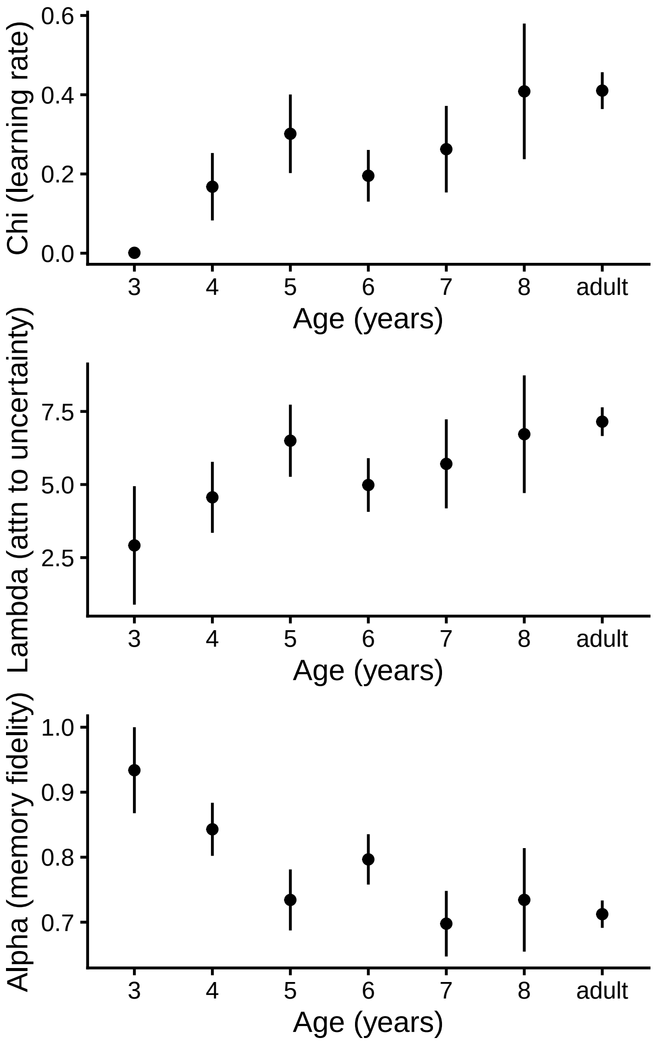

Figure 3: Participants' best-fitting model parameters by age reveal developmental change: learning rate (chi) and attention to uncertainty (lambda) generally increase with age, while memory fidelity (alpha) decreases. Children show larger individual differences, but on average match adults' values by age 8. Error bars show +/-1SE.

Selection strategies

The most likely selection strategy for each participant was calculated, and the results are shown in Table 3. Entropy was the most common strategy (60% of participants), followed by margin sampling (18%), confirmatory (12%), random (8%), and novelty (3%). Figure 4 shows the proportion of participants by age group that are most likely to be using each selection strategy. Children were more likely to use entropy maximization (84% vs. 37% of adults), whereas adults were more likely to use a confirmatory strategy (22% vs. 3% of children) or margin sampling (22% vs. 14% of children). Random selection was most likely for only 13% of adults and only 3% of children, and novelty was most likely for 5% of adults and no children.

strategy	n	proportion
entropy	114	0.60
margin	34	0.18
confirmatory	23	0.12
random	15	0.08
novelty	5	0.03

Table 3: Most likely selection strategies.

Relative contributions of strategy, age, and learning mechanisms

To assess the relative importance of sampling strategy versus learning mechanisms in predicting performance, we fitted a series of nested regression models. Sampling strategy was treatment-coded with confirmatory sampling as the baseline. A model including only sampling strategy explained minimal variance in test accuracy ($R^2 = .02$, F(4,185) = 0.94, $p = 0.44$), demonstrating that strategy choice alone is a weak predictor of learning success. In contrast, a full model including strategy, age, and learning parameters (χ, λ, α) explained substantial variance ($R^2 = 0.58$, $F(8,180) = 30.59$, $p < .001$). Learning rate (χ: $\beta = 0.33$, $p < .001$), memory decay (α: $\beta = -0.65$, $p < .001$), and attention to uncertainty (λ: $\beta = 0.01$, $p = .002$) were all significant predictors, as was age ($\beta = 0.08$, $p = .002$). Notably, strategy effects emerged when controlling for individual differences in learning mechanisms: entropy maximization ($\beta = 0.14$, $p = .007$) and margin sampling ($\beta = 0.14$, $p = .02$) significantly outperformed confirmatory sampling. These findings indicate that developmental improvements in word learning operate through measurable changes in learning rate and memory, additional age-related factors not captured by these parameters, and to a lesser extent through strategic information-seeking choices.

Discussion

We investigated the computational mechanisms underlying active word learning by fitting associative learning models to individual participants' performance and analyzing their information-seeking strategies. Two models with uncertainty-biased attention best explained learning for nearly all participants, with an uncertainty-only model (UAM) best accounting for 51% of individuals and the combined familiarity-uncertainty model (FUAM) accounting for 48%. This suggests that directing attention toward uncertain associations (i.e., items with diffuse mappings) is a fundamental mechanism in cross-situational word learning in both adults and children. These two models – differing only in whether the current association strength is considered in the update rule – very nearly mimicked each other for the bulk of the participants, but each model best explained a substantial group of participants, suggesting some individual variability in learning styles. Examining the best-fitting parameters showed that learning rate and memory decay predicted performance more strongly than attention biases. Participants

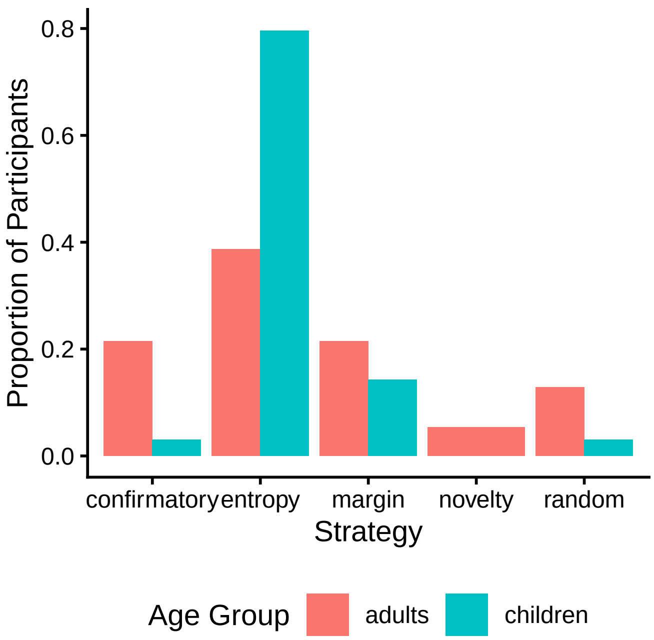

Figure 4: Best-fitting selection strategies by age group. A higher proportion of children (vs. adults) used a confirmatory strategy, while a higher proportion of adults used an entropy-based information gain strategy.

with higher learning rates (χ) and lower memory fidelity (α; effectively forgetting early ambiguous evidence) were more accurate at test. A relatively weak relationship between λ (attention to uncertainty) and performance ($r = 0.23$) suggests that how much learners encode matters more than what they attend to, at least in these experimental paradigms.

An analysis of selection strategies showed some developmental differences, but also some consistency: Adults and children both predominantly employed entropy maximization (39% of adults; 84% of children), actively seeking information about ambiguous items to reduce uncertainty. However, adults deployed more diverse strategies: 22% used confirmatory sampling (vs. 3% of children), 22% used margin sampling (vs. 14% of children), and 13% used random selection (vs. 3% of children).

This pattern suggests that children robustly deploy uncertainty-reduction strategies, consistent with broader findings on the development of information-seeking in childhood (Meder et al., 2021; Nussenbaum et al., 2019). The high prevalence of entropy maximization in children aligns with evidence that young learners actively seek information to resolve uncertainty across multiple domains (Liquin & Gopnik, 2022). However, adults' greater strategic diversity suggests that development may involve not the emergence of uncertainty reduction *per se*, but rather the acquisition of a more flexible repertoire of strategies that can be adaptively deployed depending on task demands (Ruggeri, Walker, Lombrozo, & Gopnik, 2021). This flexibility may reflect adults' greater metacognitive awareness of when uncertainty reduction is optimal versus when alternative strategies (e.g., confirmation, exploitation) are more efficient.

Why might children show less strategic diversity than adults, relying so heavily on entropy maximization? One possibility is that children's nascent metacognitive skills make it harder to flexibly shift between different information-seeking strategies based on task context (Baer & Kidd, 2022; Eccher, Mundry, & Mani, 2024). Entropy maximization may be a robust default strategy that works well across many learning contexts and requires only tracking which items are most ambiguous. In contrast, deploying confirmatory or margin sampling strategies effectively may require greater metacognitive sophistication: recognizing when one's current strongest hypothesis is worth testing (confirmation) or when near-threshold distinctions matter most (margin sampling). Thus, adults' strategic diversity may reflect the development of metacognitive flexibility that allows adaptive strategy selection based on learning goals and task structure (Ruggeri et al., 2021). Alternatively, children's limited working memory may make it difficult to simultaneously maintain multiple competing hypotheses and evaluate the diagnosticity of different sampling strategies, leading them to default to a single reliable approach (Zettersten, Choi, Kirkorian, & Saffran, 2025).

A central theoretical question is whether sampling ambiguous items causes better learning, or whether successful learning enables more sophisticated sampling strategies. Our results support the latter interpretation. Learning parameters χ and α strongly predicted performance, while sampling strategy showed weaker relationships to outcomes. This finding aligns with Zettersten & Saffran (2021)'s supplementary analysis which suggested that successfully learning novel words during training makes it more likely that learners will subsequently focus on ambiguous items. We extend this result by showing that individual differences in learning mechanisms (learning rate, memory) are the primary drivers of both performance and sampling sophistication. However, the current experimental design limits our ability to test causality: selection trials occurred *after* initial learning, and we modeled them using associations built during passive training. Future work using trial-by-trial joint optimization of learning and selection parameters could better disentangle these relationships.

Combining computational modeling of learning mechanisms with analysis of selection strategies reveals a nuanced picture of active word learning across development. Successful learning depends more on fundamental learning parameters (i.e., how quickly associations form and how readily early evidence is forgotten) rather than a bias to attend more or less to uncertain (or already-familiar) items. However, the likelihood of strategically sampling uncertain information emerges across childhood, which may suggest an increasing awareness of uncertainty, or of the value of reducing it. These findings advance our understanding of how learners actively shape their own language learning experience, and how cognitive development enables increasingly sophisticated information-seeking strategies.

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